Directionality Characteristics of the Tympan Open-Source Hearing Aid and Earpieces

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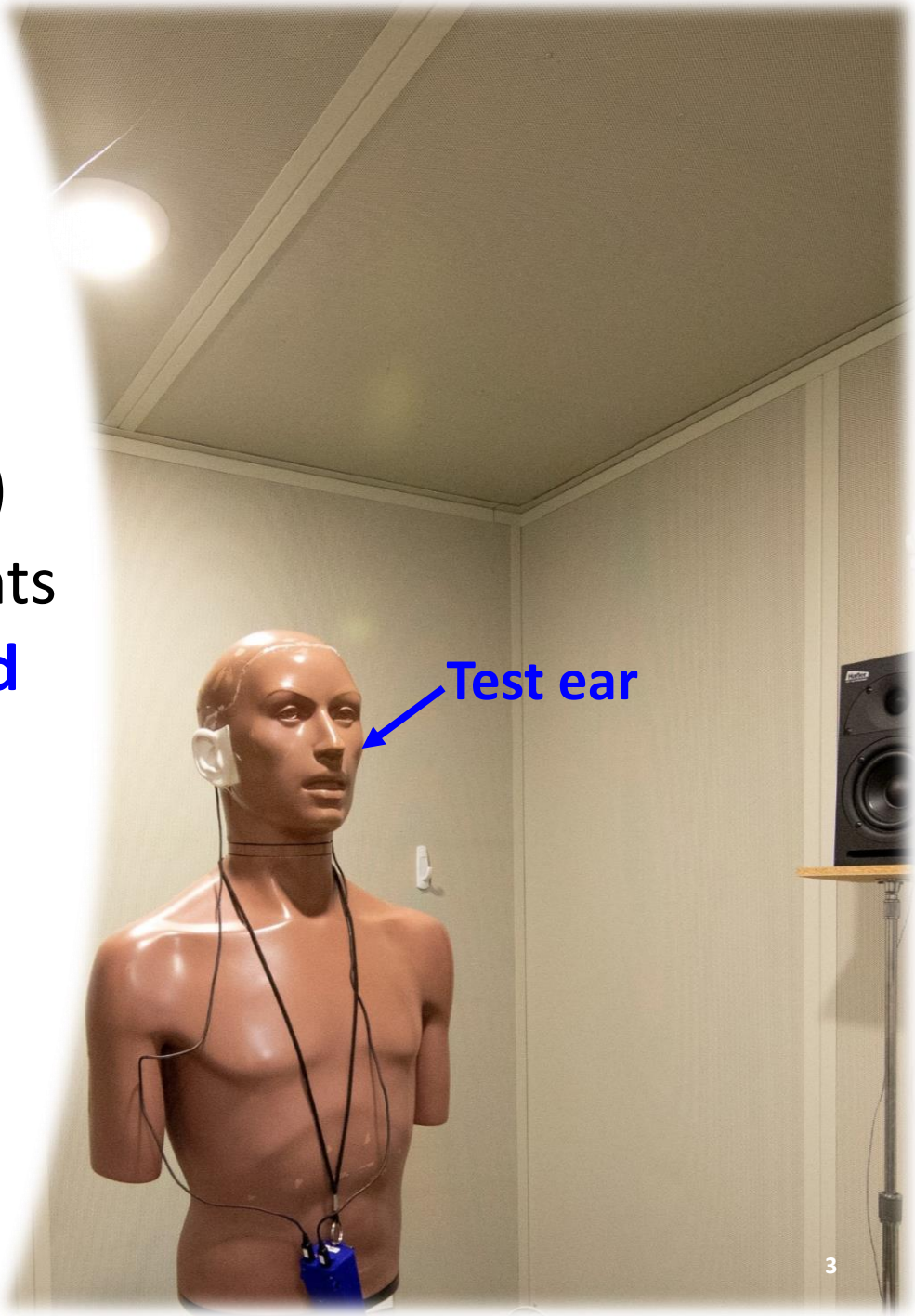
Methods

- When possible, followed procedures outlined in **ANSI/ASA S3.35-2021**
  - *"Method of measurement of performance characteristics of hearing aids under simulated real-ear working conditions"*

- **Pink Noise Stimulus**
  - ≈85 dB SPL at hearing aid mic location
  - 5-second duration
  - Analyzed in 1/3-octave bands

Set up

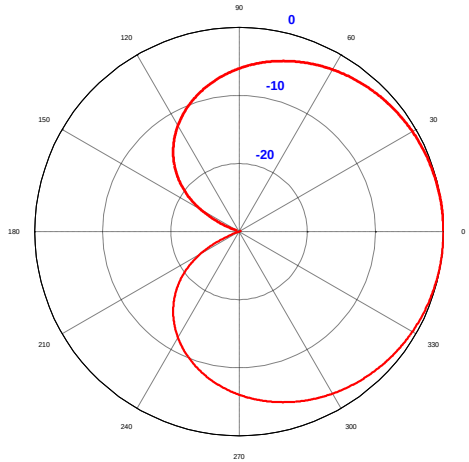
- KEMAR centered in a double-walled sound booth (2.13 m x 2.44 m)
  - Rotated in 15° increments
  - **Left earpiece was tested**
- Hafler M5 Reference Monitor sound field loudspeaker
  - Flat frequency response



Earpieces

- 9.5-mm mic spacing (d)
  - c ≈ 343 m/s
  - Travel time (d/c): 27.7 µs

- Sampling rate set to 36.1 kHz → 1-sample delay = d/c = 27.7 µs
  - Cardioid directivity



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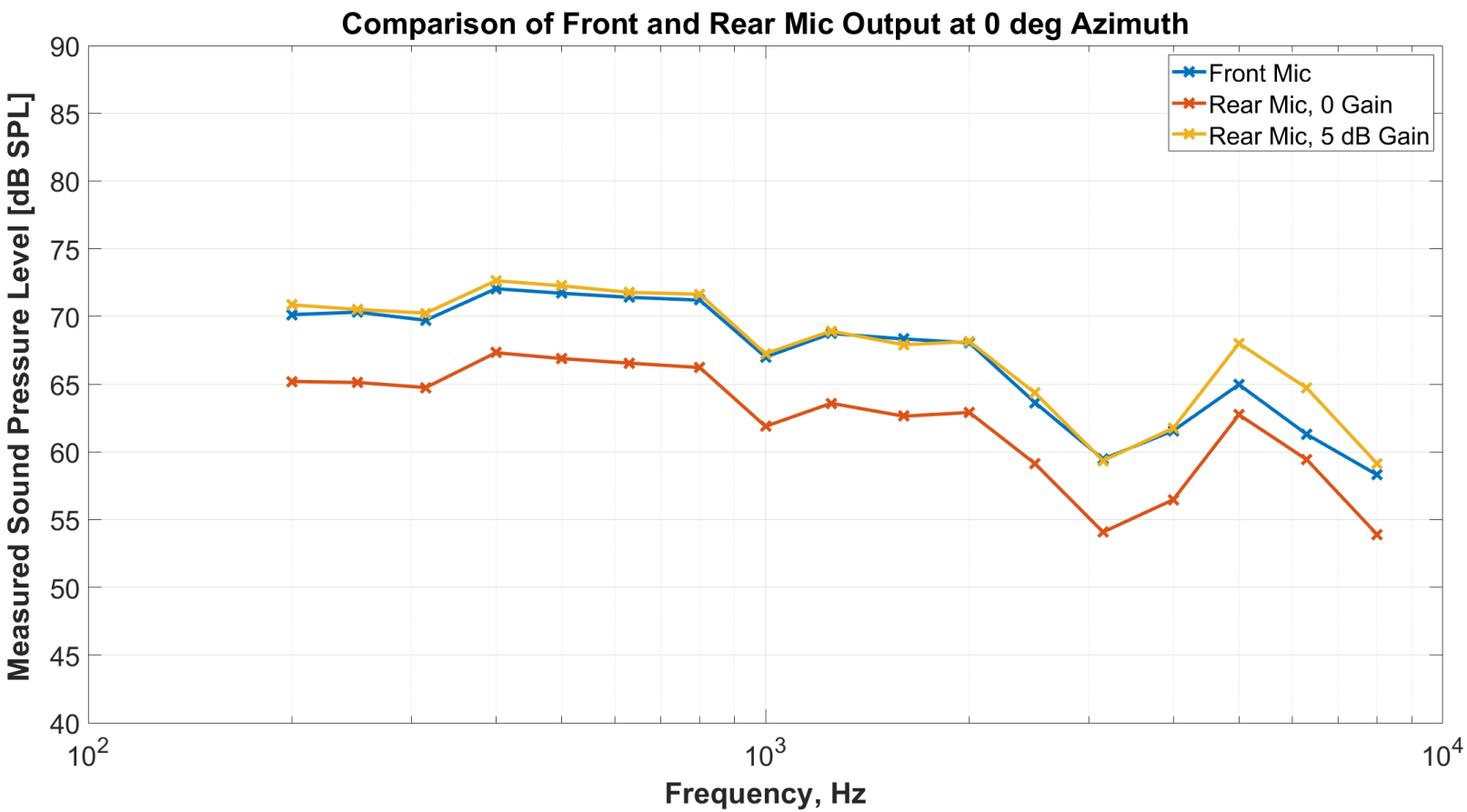
Microphone Matching

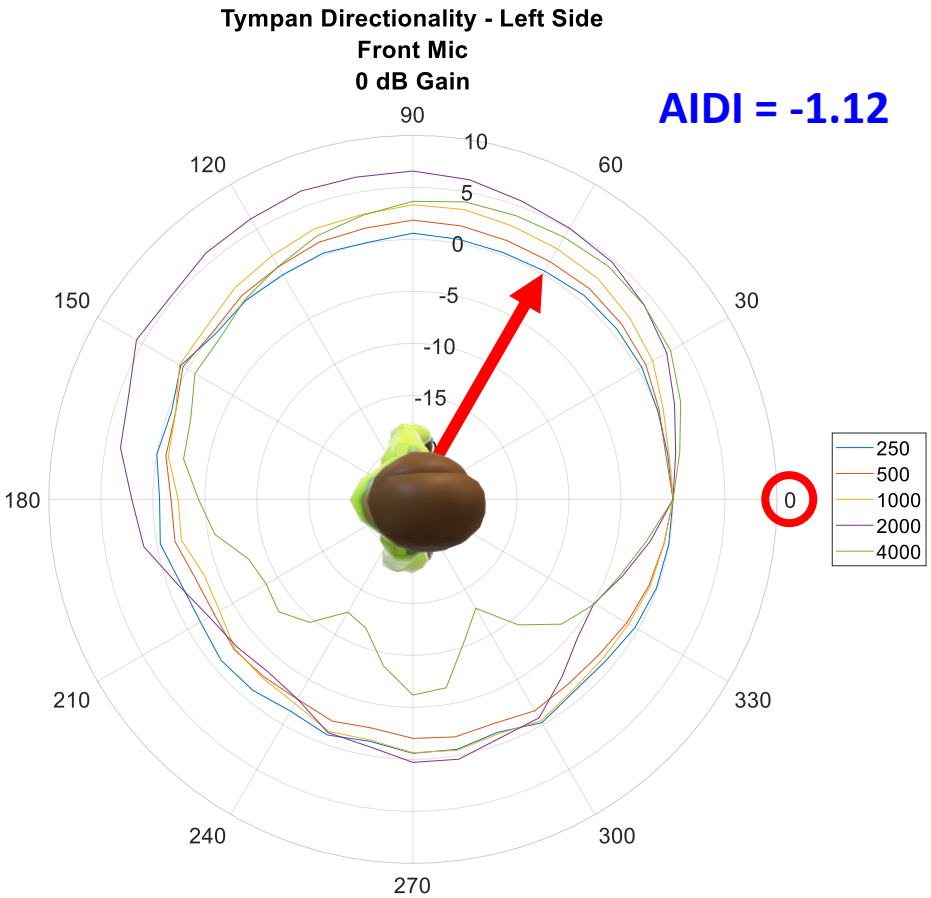
- Critical for optimal cancellation
  - **Rear mic output** was ≈ 5 dB lower than the **front mic**
    - 5 dB of gain was added to **match the two mics**



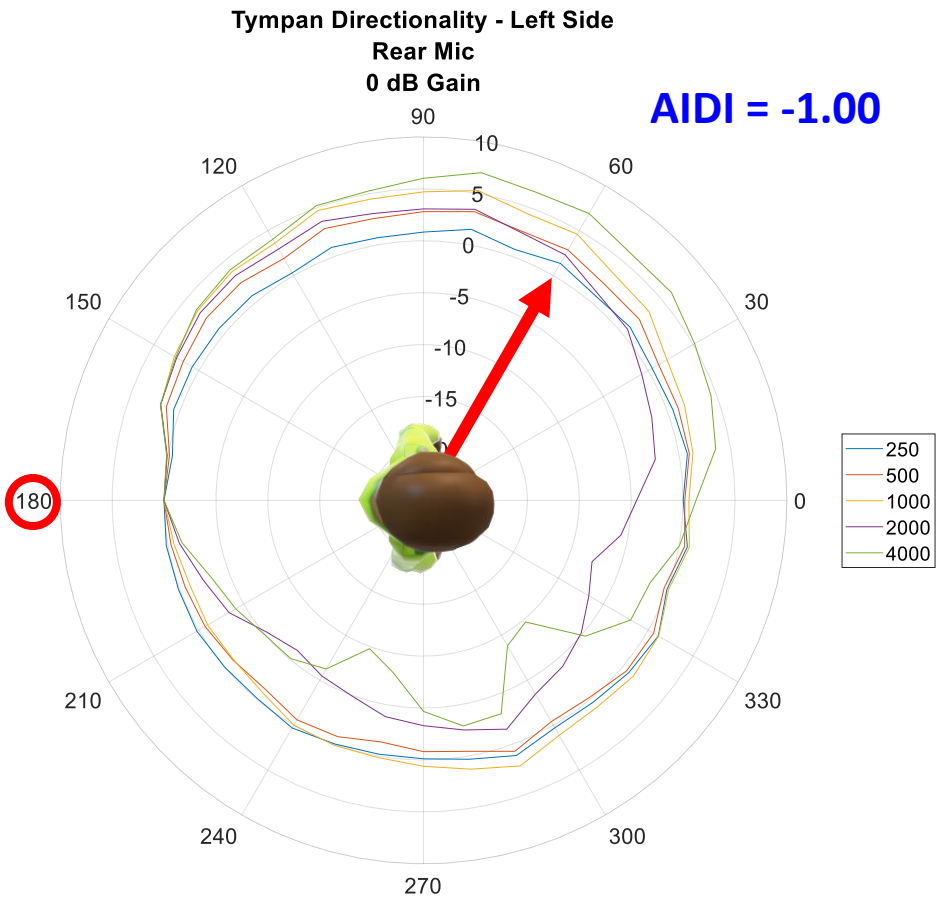
Omni-Directional Responses

- It is well-known that putting mics on the head will shift the response to the side due to head shadow

Front Mic (re 0°)



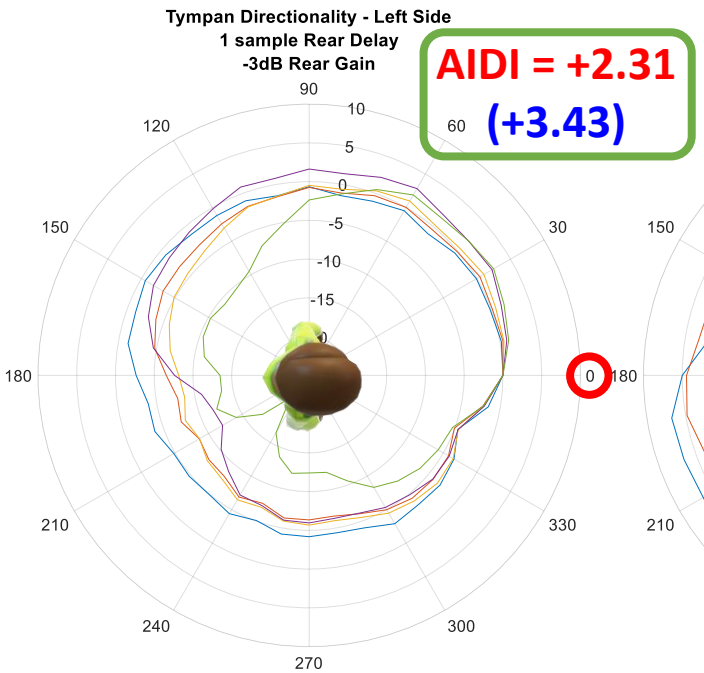
Rear Mic (re 180°)



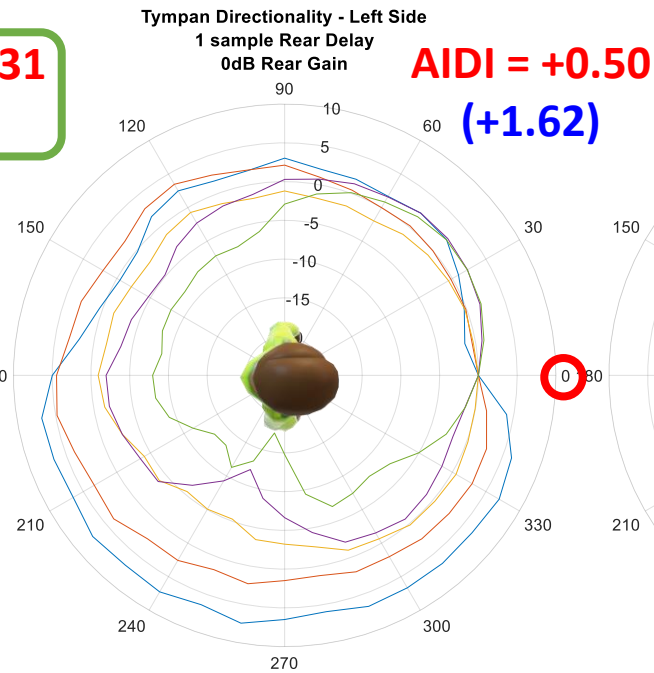
Rear Mic Delay

0° reference

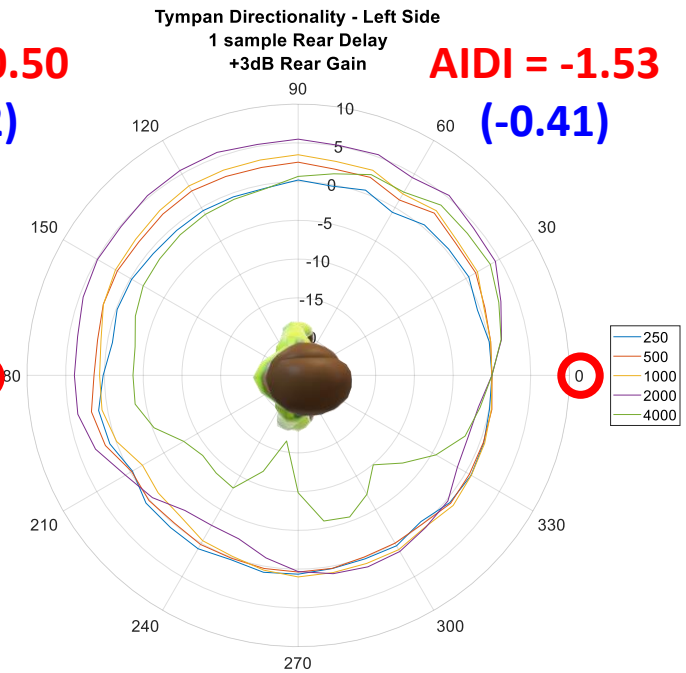
-3 dB Rear Gain



0 dB Rear Gain



+3 dB Rear Gain



*Gain values after mic matching

AIDI was lowest with -6 dB and +6 dB rear mic gain

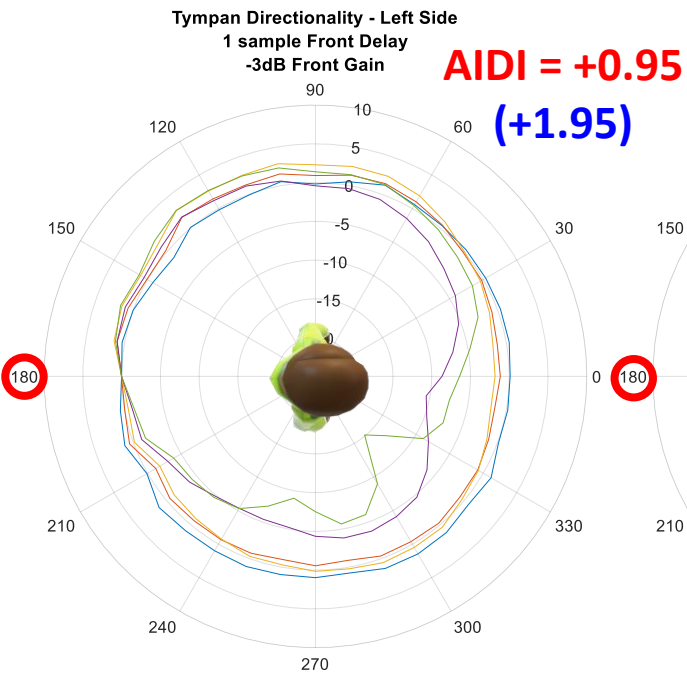
AIDI = Articulation Index weighted Directivity Index

values in () = **Improvement** re: front mic only

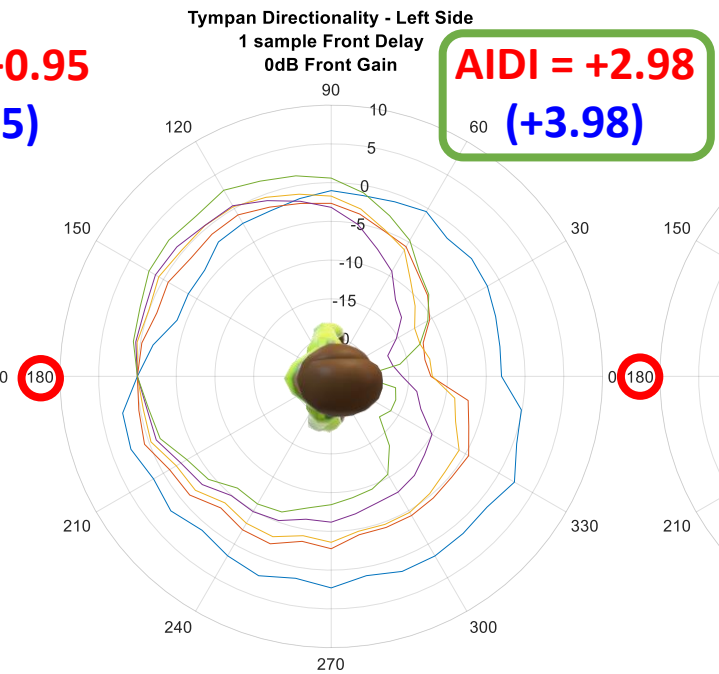
Front Mic Delay

180° reference

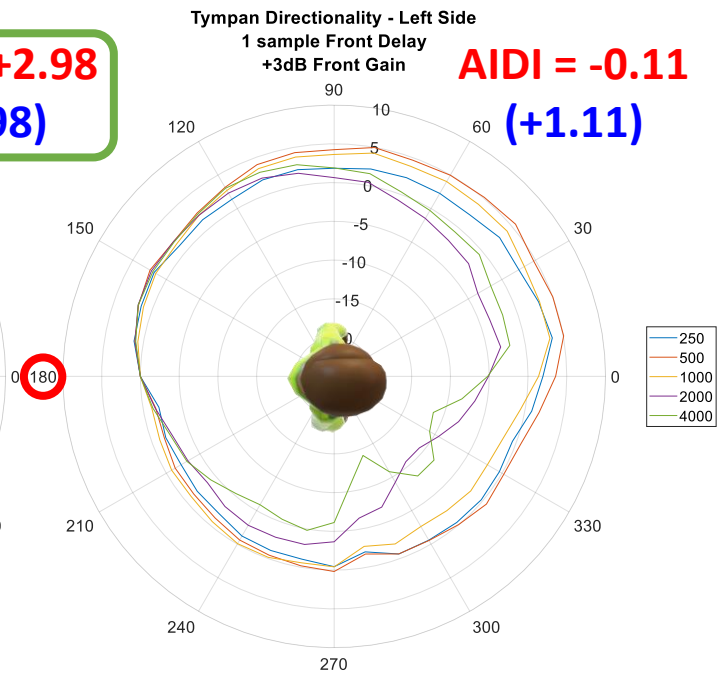
-3 dB Front Gain



0 dB Front Gain



+3 dB Front Gain



*Gain values after mic matching

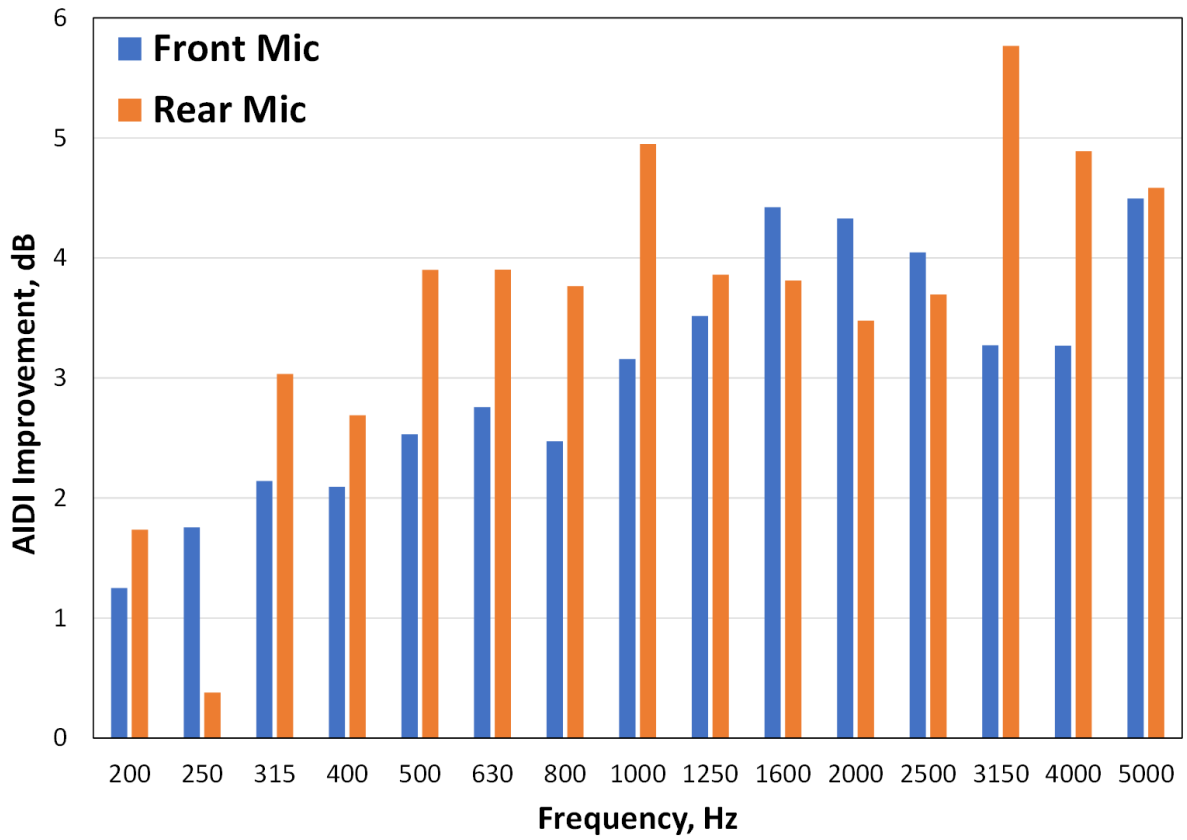
AIDI was lowest with -6 dB and +6 dB front mic gain

AIDI = Articulation Index weighted Directivity Index

values in () = **Improvement** re: rear mic only

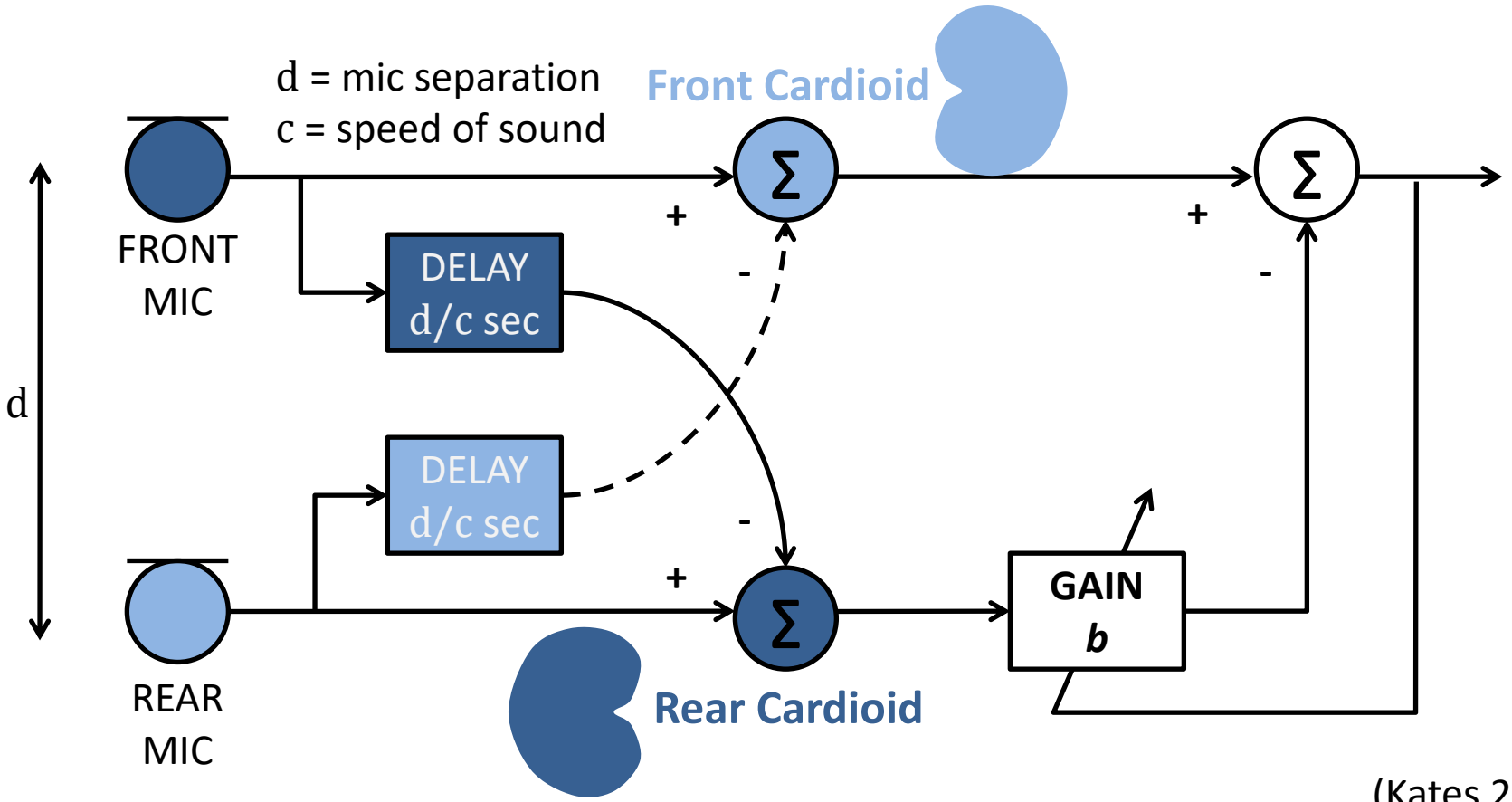
AIDI Improvement

- Front mic with rear mic delay and gain = -3 compared to front mic only (0° reference)
- Rear mic with front mic delay and gain = 0 compared to rear mic only (180° reference)



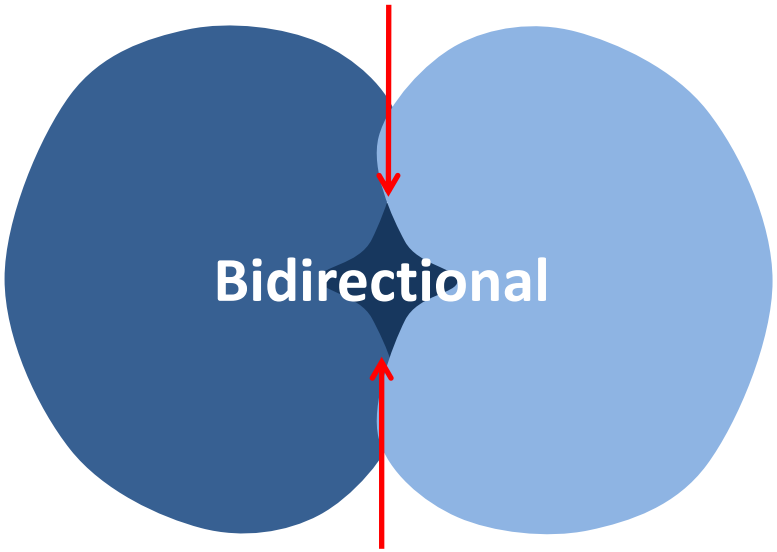
Future Directions: Variable Directionality

- **Step 1:** create opposing cardioid patterns
  - Front facing
  - **Rear facing: inverted phase and variable gain, *b***



(Kates 2008)

Future Directions: Variable Directionality



***b* = 1**

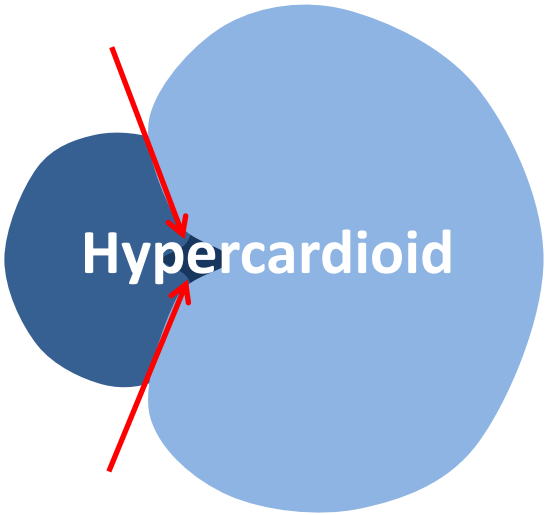
- **Step 2:** vary relative gain *b* on rear cardioid to create different polar patterns with a continuous range of nulls

Gain on Rear Mic (*b*)

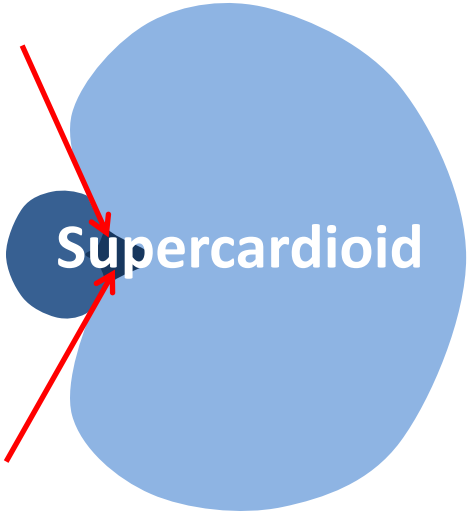
0.0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0
180° (cardioid)		130° 230°			110° 250°					90° 270°

Azimuths of Nulls

The **front** and **rear** responses **add with opposite sign**, thereby creating nulls where they intersect



***b* = 0.5**



***b* = 0.25**

Conclusions

1. Optimized directionality requires **precise timing** (sampling rate) and **microphone matching**

2. Directional patterns and relative AIDI improvements (≈3-4 dB) are **consistent with published research**

3. **Optimal directivity**
   - Front cardioid obtained with -3 dB gain on the rear mic
   - Rear cardioid obtained with matched mic gain

4. **Future directions:** use optimized front and rear cardioids with opposite phase to **vary null with fixed and adaptive directionality**